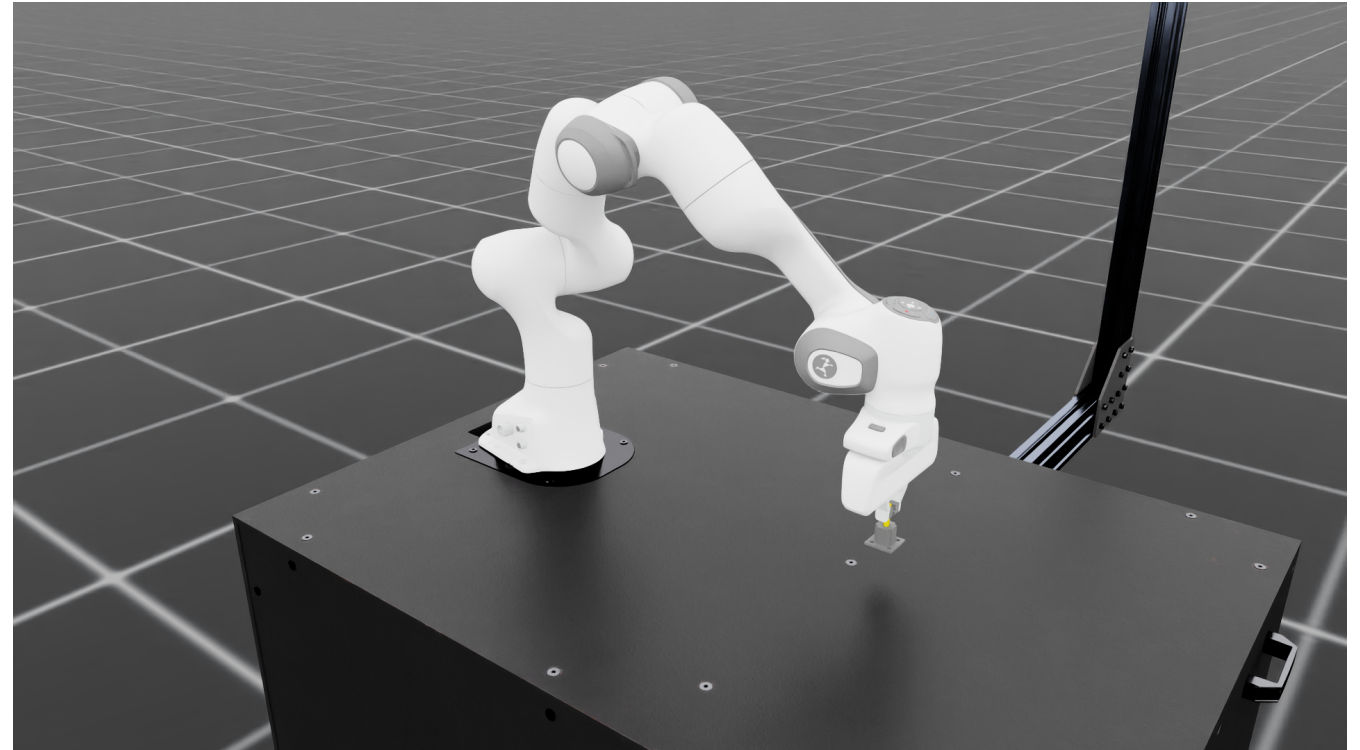


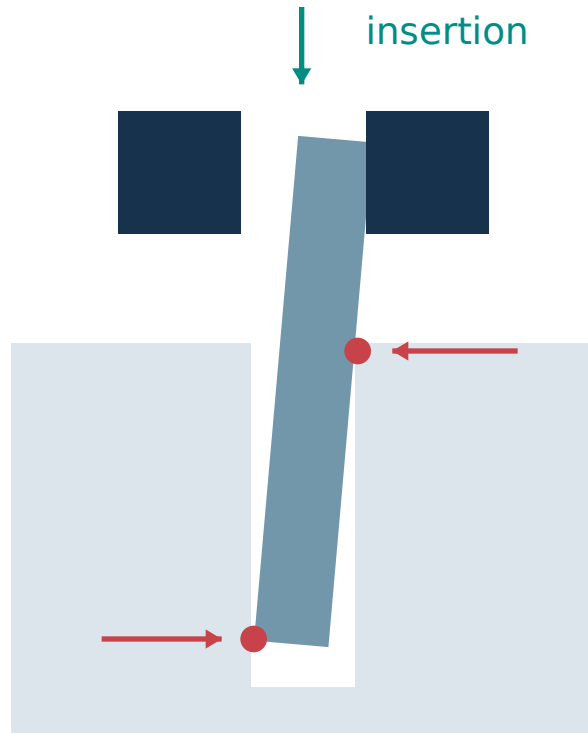
Preventing insertion-induced jamming

First avoid risky insertion.
Then restore a force-limited exit.

Research progress and technical roadmap
September 2026



A tight insertion can create a difficult exit



Metal peg / tight plastic socket
Possible opposing contacts

01 Alignment error

Tilt and deep engagement can create opposing edge or wall contacts.

02 Restricted motion

Contact reactions and friction can resist both insertion and withdrawal.

03 Change the contact state

Lateral or angular correction may be needed before a low-force exit is possible.

Why pulling harder is not a general solution

1 Grasp capacity

Pull requires fingertip friction.



More pull requires more squeeze.



Slip or local damage limits the grip.

2 Part integrity

Tilt concentrates contact loading.



Friction and bending load the socket.



Plastic, surfaces or fixtures may fail.

3 Controlled release

High load deflects the assembly.



Elastic energy can accumulate.



Sudden release can cause rapid motion.

Ideal friction grip: $F_{\text{pull}} \leq \mu_g (N_L + N_R)$

N_L, N_R : finger normal forces. μ_g : grip friction.

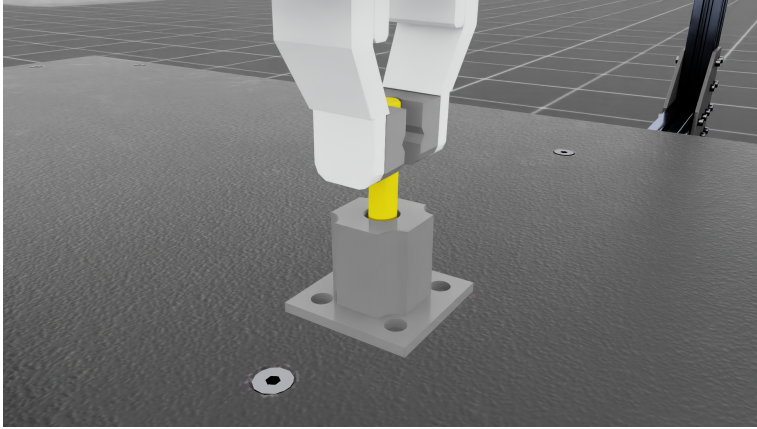
Current proxy: wrist norm $\leq B$, torque $\leq 1 \text{ N}\cdot\text{m}$.
3 / 4 / 5 N are exploratory budgets, not calibrated limits.

Research context

Contact and extraction	Geometry, friction and compliance shape mating forces.	Whitney 1982 [1] Goli et al. 2024 [2]
Interaction control	Virtual stiffness and damping shape contact response.	Hogan 1985 [3]
Planning through contact	Trajectory optimization can include changing contact modes.	Posa et al. 2014 [4]
Safety and transfer	Predictive filtering and calibrated deployment address model risk.	Wabersich 2021 [8] IndustReal 2023 [6]

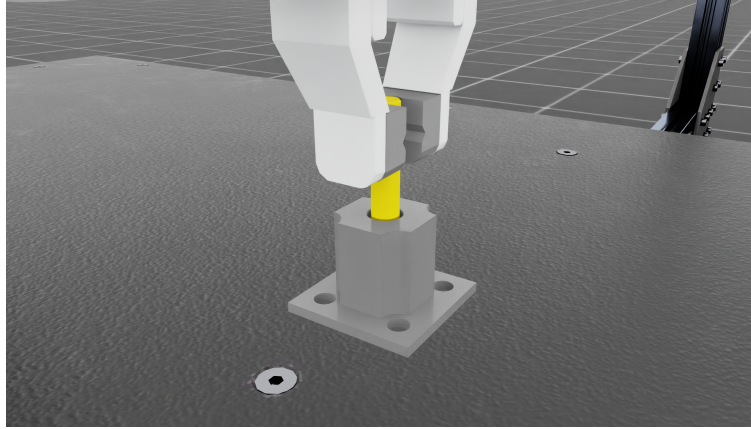
First avoid risky insertion. Then restore an exit within the force budget.

Current testbed and recovery probes



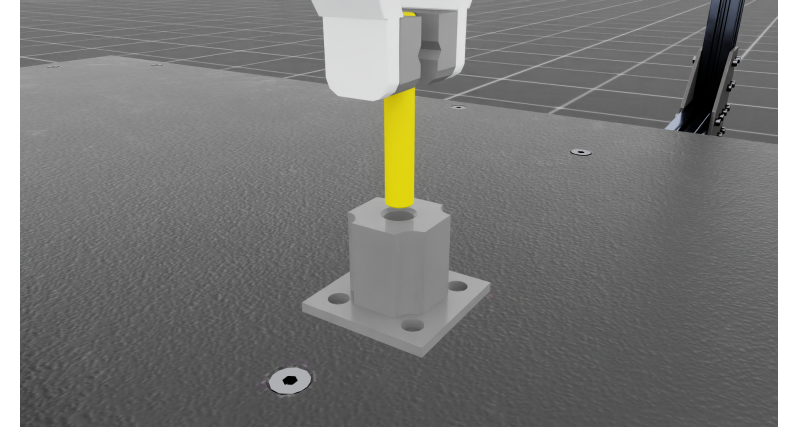
Reference state

18.228 mm depth, 1.439° tilt



Recentered and realigned

18.076 mm depth, 0.170° tilt



Clearance

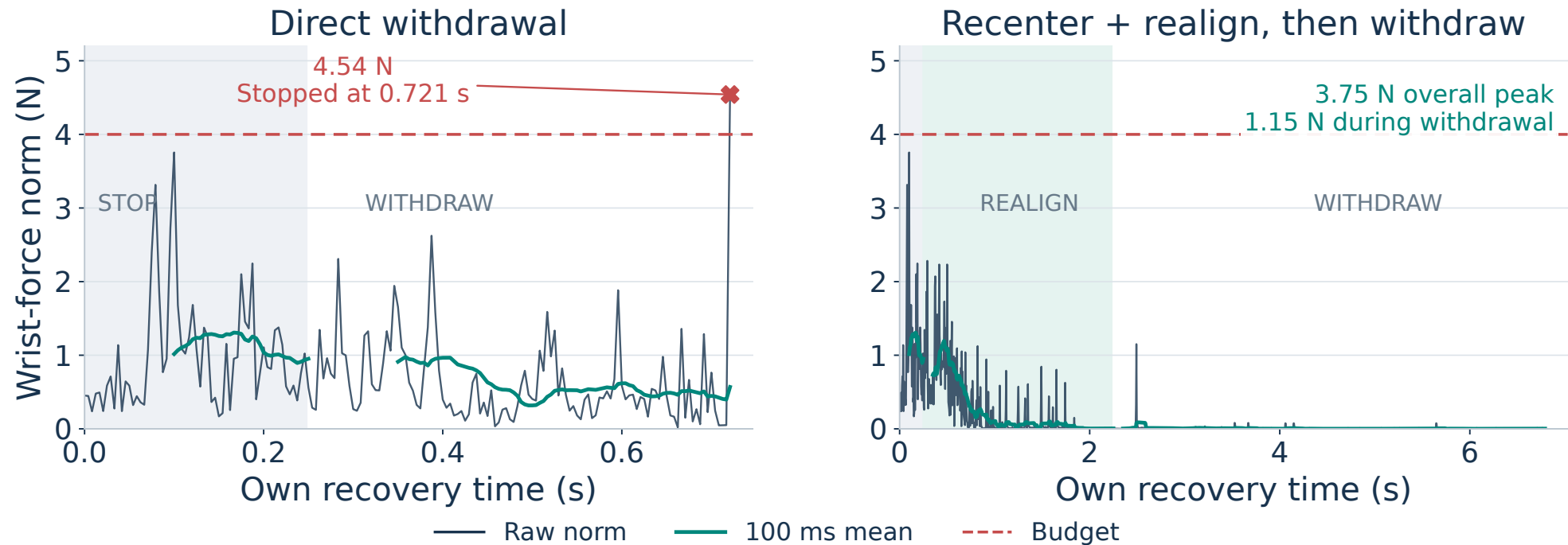
-3.009 mm depth, 0.063° tilt



Rigid bodies, 0.2 mm radial gap, 25 mm blind hole. Physics and task control: 240 Hz.

Evidence: the exit depends on the motion

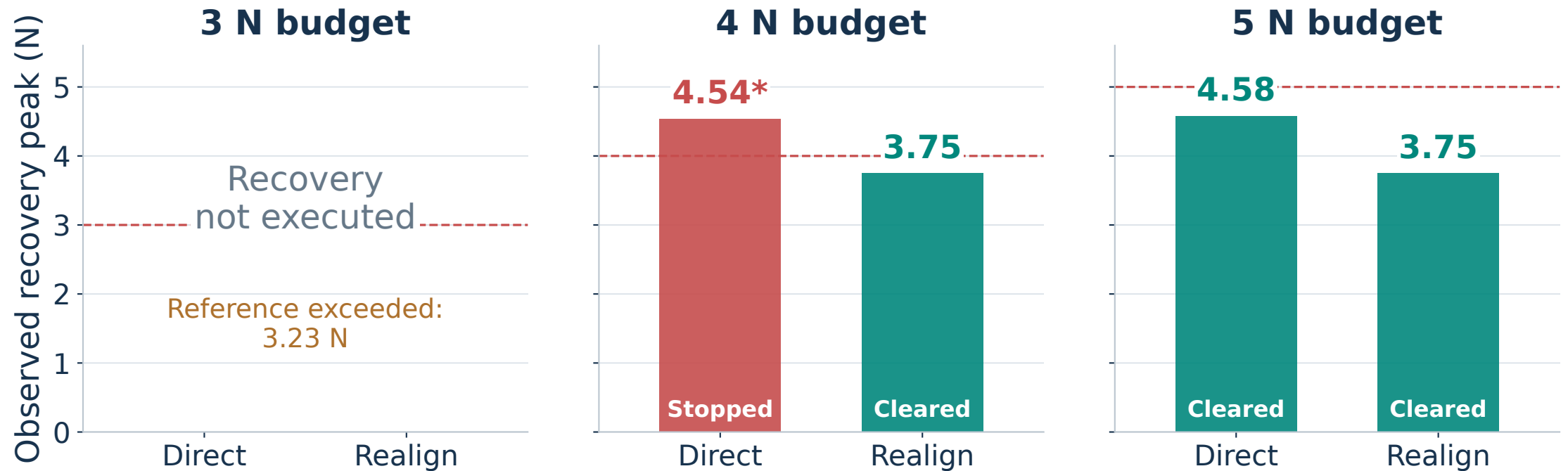
18 mm target depth · $\mu = 1$ · 1.5° command tilt · 4 N budget



Matched reference: 2,811 samples per policy · reference peak 3.56 N.

Direct path stops at first exceedance. This pair motivates prevention; it does not validate it.

The budget changes feasibility



* First-exceedance stop; complete direct-withdrawal peak remains unobserved.

Recoverability depends on state, candidate policy and the task budget.

Learn when insertion becomes costly to undo



Why this helps

A state may still permit insertion while making withdrawal expensive. Predict that cost before committing to the next insertion segment.

Start simple

Rules or a small classifier → temporal model / ensemble only after collecting enough matched histories.

Keep censored and untested outcomes separate. Split data by shared physical path.

Primary output: an early warning that changes the next insertion action.

Prevent jamming before advancing deeper

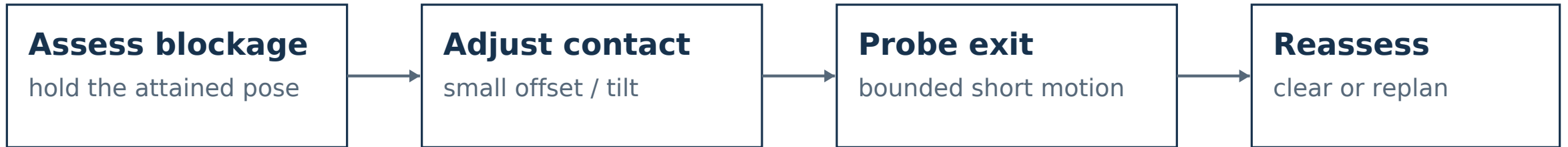


Objective: insertion progress – risk penalty – motion effort

- PLAN [4,8]** Local MPC selects depth, offset, tilt and speed while retaining a predicted exit.
- CONTROL [3]** Use Jacobian-based Cartesian control; tune lateral/rotational stiffness and damping.
- FILTER [8]** Modify or reject a reference if predicted load exceeds B minus an uncertainty margin.

Rules → local model predictive control (MPC). Compliance alone does not enforce the force limit.

Restore withdrawability under the same limit



Search the entire recovery path

Choose the candidate with the lowest predicted peak load that reaches clearance and retains the grasp.

How it works

Change contact geometry before the full axial pull.
Include adjustment cost, uncertainty and stopping response.

Begin with a motion library. Add local MPC and predictive filtering after model validation.

No admissible candidate → no validated action in the tested library; reassess the task.

Validate the mechanism, then the policy

01 Trust the measurements

Separate physics and control rates.
Check solver, speed and load-signal sensitivity.

02 Test prevention

Baseline vs risk-aware insertion under matched disturbances.
Report completion, peak load, aborts and later exit success.

03 Test recovery

Matched blocked states, same force budget.
Report completion, peak load, overshoot and grasp retention.

04 Calibrate and transfer

Measure grip and part/fixture limits.
Randomize identified uncertainty before hardware tests. [6]

Control barrier functions (CBFs) [5]: a later option after validating dynamics and backup actions.

Prevent first. Recover within limits if needed.

What we have

**A reproducible adverse scenario:
direct exit exceeds 4 N; a different motion clears.**

Primary research

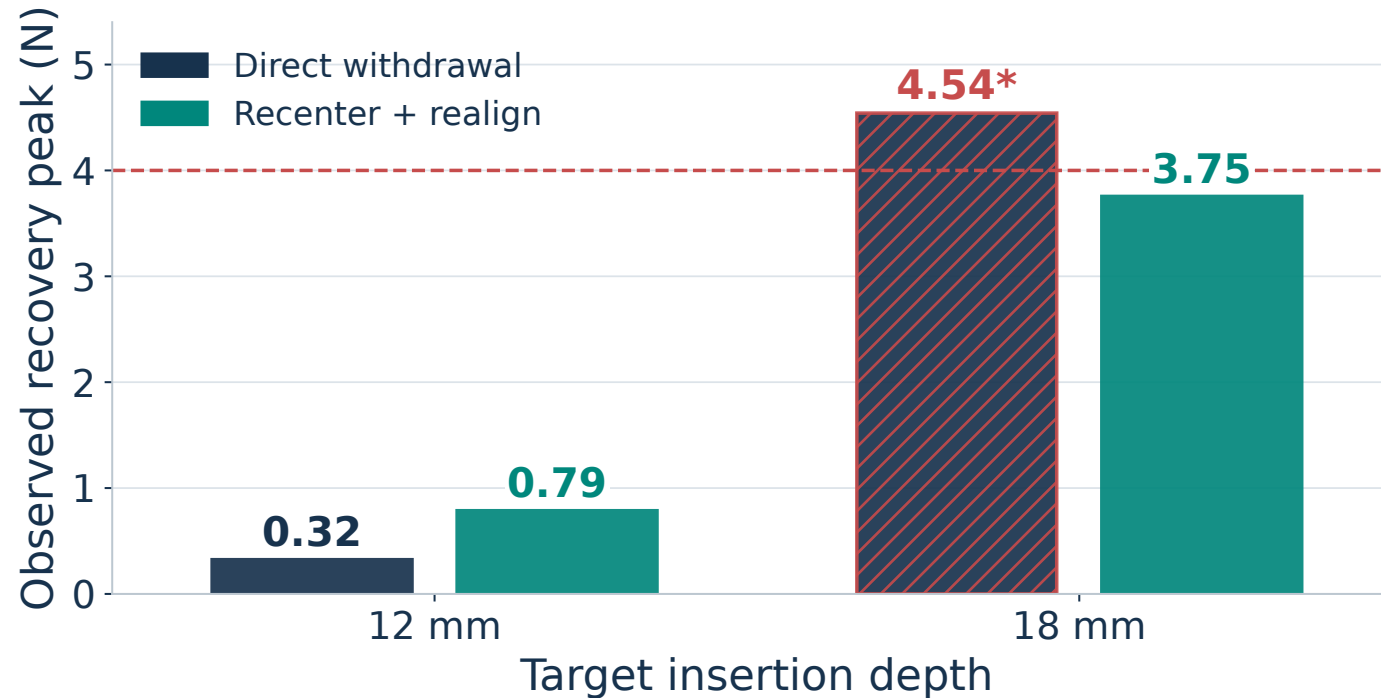
Predict risk from insertion history.
Choose motions that avoid jam-prone states.

Secondary research

Restore a feasible exit when progress is blocked.
Constrain both adjustment and withdrawal.

Recovery scenario established. Preventive insertion control is the next test.

Embedding depth changes the recovery cost



* Censored at budget exceedance. Actual initial tilt: 1.52° / 1.44°.

18 mm direct path

1.105 mm

Commanded withdrawal
at the stopping frame

0.137 mm

Measured withdrawal
at the same frame

Recovery cost can inform insertion risk. Prevention needs closed-loop tests.

Budget-study evidence coverage

Depth	μ	Tilt	Budget	Direct withdrawal	Recenter + realign
18 mm	1	0°	4 N	Clear within budget	Clear within budget
18 mm	1	1°	4 N	Clear within budget	Clear within budget
18 mm	1	1.5°	4 N	Exceeded in withdrawal	Clear within budget
18 mm	1	1.5°	5 N	Clear within budget	Clear within budget
12 mm	1	1.5°	4 N	Clear within budget	Clear within budget
18 mm	0.5	1.5°	4 N	Exceeded in reference	Not tested
18 mm	1	2°	4 N	Exceeded in reference	Not tested
18 mm	1	1.5°	3 N	Exceeded in reference	Not tested

[D1] Budget study: 8 conditions, 13 executed branches, 3 untested.

Exploratory deterministic trials. Shared histories and repeated anchors are not independent samples.

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